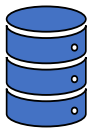
Vehicle Claims and Performance Analysis



📌 PROJECT OVERVIEW

This project is an end-to-end **Power BI business analytics solution** developed to analyze a simulated vehicle service and warranty claims environment. The primary objective was to transform raw transactional data into a **decision-ready analytical dashboard** that helps stakeholders understand **claim activity, financial exposure, claim severity, approval and denial behavior, call-center performance, dealer performance, vehicle patterns, mileage exposure, and operational processing efficiency**.

Rather than treating the assignment as a simple reporting exercise, the analysis was structured around a broader business question: **Where is the financial and operational impact concentrated, what patterns or anomalies stand out, and what relationships may explain those outcomes?** The final solution therefore combines **data preparation, dimensional modeling, DAX, interactive visualization, trend analysis, anomaly investigation and business recommendations**.



📁 DATASET DETAILS

The dataset contains **5,000 records across 22 original fields**, with each row representing **one vehicle service or warranty call/claim interaction**. The data covers activity from **January 2020 through December 2022**, while claim decisions extend into **2023**.

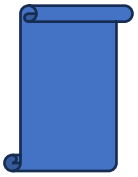
The dataset includes information across several business dimensions, including **call details, contract information, claim characteristics, vehicle information, dealer information, financial amounts, employee performance and processing times**.

Key dimensions represented in the data include **50 agents, 20 selling dealers, 20 servicing dealers, 5 vehicle makes, 5 claim types, 2 call types, 2 contract types, and 2 claim decisions**.

The major claim categories are **Electrical, Engine, Suspension, Tires, and Transmission**, while the vehicle makes are **BMW, Honda, Hyundai, Kia, and Toyota**. The financial component of the dataset represents approximately **$25.04M in total claim exposure**, providing enough scale for meaningful financial and operational analysis.

	Field	Suggested Data Type	Data Nature	Classification
1	Call ID	Text	Identifier	Qualitative/Categorical
2	Date	Date	Time-based	Quantitative
3	Time	Time	Time-based	Quantitative
4	Agent	Text	Category	Qualitative/ Categorical
5	Call Type	Text	Category	Qualitative/ Categorical
6	Contract No	Text	Identifier	Qualitative/ Categorical
7	Selling Dealer	Text	Category/Identifier	Qualitative/ Categorical
8	Service Dealer	Text	Category / Identifier	Qualitative/ Categorical
9	Claim Type	Text	Category	Qualitative / Categorical
10	Miles Since Purchased	Whole Number	Numeric measurement	Quantitative/ Continuous
11	Contract Type	Text	Category	Qualitative / Categorical
12	Vehicle Make	Text	Category	Qualitative / Categorical
13	Claim Amount	Currency / Decimal Number	Numeric	Quantitative/ Continuous
14	Claim Decision	Text	Category	Qualitative / Categorical
15	Claim Decision Date	Date	Time-based	Quantitative
16	Talk Time Mins	Decimal Number	Numeric measurement	Quantitative/ Continuous
17	Hold Time Mins	Decimal Number	Numeric measurement	Quantitative/ Continuous
18	Handle Time Mins	Decimal Number	Numeric measurement	Quantitative / Continuous

	Field	Suggested Data Type	Data Nature	Classification
19	After Call Work Mins	Decimal Number	Numeric measurement	Quantitative/ Continuous
20	Agent Score	Whole Number	Rating	Quantitative/Discrete
21	Year	Not directly stated as a separate column	Derived if created	Quantitative / Discrete
22	Decision Processing Time	Potential derived field	Derived	Quantitative/Continuous



🎯 THE PROBLEM STATEMENT:

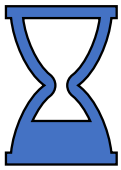
This project analyzes a **5,000-record simulated call center dataset** containing customer service and warranty interactions related to vehicle contracts and claims. The analysis is being developed in **Microsoft Power BI** to transform raw operational data into an interactive business intelligence solution that makes **trends, anomalies, performance patterns, and potential areas of improvement** easier to identify.

The core business problem was to understand **how vehicle claims behave financially and operationally and where potential areas of concern or opportunity exist**.

The analysis was designed to answer questions such as: **Which claim categories create the greatest financial exposure? Which vehicle or mileage segments stand out? Are approval and denial outcomes consistent? Where do dealers differ in financial exposure or processing efficiency? Are there unusual periods in the data that deserve investigation?**

The broader objective was therefore not simply to create charts, but to develop a dashboard that supports **observation → evidence → relationship → business implication**.

This approach helped ensure that each major visual answered a business question and contributed to an overall analytical story.



🔎 DATA EXPLORATION AND CLEANING

The project began with a structured **data exploration and Power Query transformation process**. The raw fields were reviewed to understand their meaning, granularity, cardinality, data quality, and analytical purpose before visualization.

The dataset contained **no null values**, and Call_ID was confirmed to be unique across all 5,000 records. Identifier fields such as **Call ID, Contract Number, Selling Dealer and Servicing Dealer** were therefore treated as text because they represent identifiers rather than quantities that should be aggregated.

Date fields were converted to appropriate **Date** types, while Time was retained as a **Time** field rather than converting it into text. Numeric fields such as **mileage, claim amount, talk time, hold time, handle time, and after-call work time** were assigned appropriate numerical data types.

A separate numeric field, **Agent_Score_Value**, was created from the original descriptive score values such as 1 - Bad through 5 - Great, allowing the score to be analyzed quantitatively while preserving the original descriptive field.

Initially, I believed that it is necessary, thus, A **Mileage_Band** was also created to segment vehicles into **0–10K, 10K–25K, 25K–50K, 50K–75K, and 75K+** categories but as analysis moved further, I didn't processed to make the Dashboards Compact but this field would be a measure further. Additional analytical fields were created, including **Decision_Processing_Days**, derived from the difference between claim decision date and original claim date, and **Call_Hour**, extracted from the time field. The cleaning process also included validation of internal relationships. For example, **Handle Time was confirmed to equal Talk Time + Hold Time** across the records and decision-processing durations were reviewed to ensure they were logically valid



🧩 WHAT'S INSIDE THE DASHBOARD

🏠 01 – Executive Overview

Provides a high-level snapshot of the portfolio through **core KPIs, claim distribution, financial exposure, approval behavior, vehicle mix, contract mix and Year & monthly movement**.

💰 **02 – Claims & Financial Overview**

Focuses on **claim Amount by claim type, financial exposure by contract type, Approved, Denied and their rates.**

🎧 **03 – Agent Performance**

Analyzes **Service vs Warranty activity, talk time, hold time, hourly activity, ratings consolidated by agents, Analyzed claim amount by claim type and contract type between each agents and a Interactive cards to determine top "Great" and top "Bad" rated agents.**

🏢 **04 – Service Dealer & Process Analysis**

Examines dealer-level **total claim exposure, average claim severity, Approved**, Denied claims by their rates with Interactive cards to determine highest and lowest service dealer as per claim amount, providing both financial and operational perspectives.

🏢 **05 – Selling Dealer & Process Analysis**

Examines dealer-level **total claim exposure, average claim severity, Approved**, Denied claims by their rates with Interactive cards to determine highest and lowest service dealer as per claim amount, providing both financial and operational perspectives.

🚨 **06 – Anomalies & Trends**

Acts as the investigative layer of the solution by examining **monthly & yearly financial movement, approval/denial changes, Contract & claim type amount across approved and denied claims with highlighting highest and lowest trends**

The page structure was intentionally designed to move from **high-level monitoring to deeper investigation**.

💰 **07 – Deep Dive analysis**

This Analysis contains all core fields with the highest and lowest slicer to examine different fields across Claim type, contract type and year.



🧮 DAX Formulas & New Measures

A reusable **DAX measure layer** was created so that calculations could be consistently applied across filters, dimensions, and visuals rather than relying on hard-coded aggregations.

Core measures included:

Total Calls, **Total Claim Amount**, **Average Claim Amount**, **Average Mileage**, **Approved Claims**, **Denied Claims**, **Approval Rate** and **Denial Rate**.

Operational measures included **Average Talk Time**, **Average Hold Time**, **Average Handle Time**, **Average After Call Work** and **Average Decision Processing Days**.

Employee-performance analysis used **Average Agent Score**, while portfolio structure was supported by measures such as **Total Agents** and processing-time minimum/maximum measures. The use of DAX measures created a **consistent analytical layer**, improved reusability, and allowed the same business logic to dynamically respond to slicers and filter context.



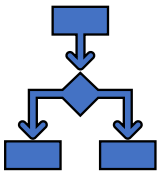
📊 VISUALS UTILIZED TO ANALYZE THE DATA

A combination of visualization types was deliberately used based on the analytical question rather than simply for presentation variety.

KPI cards were used for executive-level metrics. **Clustered bar and column charts** were used for categorical comparisons such as claim type, vehicle make, dealer, mileage band, and financial exposure. **Line charts** were used to identify **monthly trends and potential anomalies over time**. **Donut charts** were used selectively for high-level distributions such as claim decisions.

Matrices were used where multiple dimensions needed to be investigated simultaneously, including **Claim Type × Vehicle Make** and **Contract Type × Claim Type × Approved/Denied/Total**.

A particularly advanced element was the **Scatter/Bubble Chart with Play Axis**, combining **Average Claim Amount, Approval Rate, Total Claim Amount, Vehicle Make and Year-Month** into a dynamic multidimensional visual. This was intentionally included to demonstrate that the dashboard could support **interactive relationship analysis rather than only static comparisons**.



📊 STEPS I FOLLOWED TO BUILD THE DASHBOARD

The dashboard was developed through a structured end-to-end process beginning with **data ingestion and profiling**.

The data was then cleaned and standardized through **Power Query**, followed by creation of analytical fields such as **Decision Processing Days, Call Hour, Mileage Band and Agent Score Value**. A practical data model was then established using **Fact_Vehicles** supported by three focused dimensions: **Dim_Date, Dim_Agent and Dim_Vehicle**. I created a new sheet including vehicle images data and connected a relationship with Dim_vehicle

The modeling approach was deliberately selective. Rather than creating dimensions for every categorical field, the model retained low-cardinality operational categories directly within the fact table because the dataset contained **only 5,000 records and a single primary fact table**.

Next, the **DAX measure layer** was established. The report was then developed progressively, beginning with the **Executive Overview**, followed by deeper financial, operational, dealer, vehicle and anomaly-oriented analysis.

Finally, the dashboard was enhanced with **interactive filtering, comparative analysis, matrices, trend views, and the advanced Play Axis visualization**.



💡 KEY INSIGHTS

💰 Financial Exposure

The analysis shows that the dataset represents approximately **$25.04M in total claim exposure**, with financial impact distributed relatively evenly across the major claim categories. **Suspension generated the highest total claim exposure at approximately $5.20M**, while **Engine recorded the highest average claim value at approximately $5,066**.

🚗 Vehicle-Level Financial Patterns

At the vehicle level, **Honda recorded the highest total claim exposure at approximately $5.19M**, followed by BMW, Kia, Toyota, and Hyundai. However, the severity picture was different, with **Kia recording the highest average claim amount at approximately $5,108**.

🛣️ Mileage & Financial Exposure

Mileage analysis revealed a notable concentration of financial exposure in higher-mileage vehicles. The **50K–75K mileage band generated the highest total claim exposure at approximately $8.49M**, compared with only **$3.44M for the 0–10K band**.

🏢 Servicing Dealer Performance

The servicing-dealer analysis identified meaningful differences in **financial exposure and claim outcomes**. **Servicing Dealer 009 recorded the highest total claim exposure at approximately $1.40M**, while **Dealer 010 recorded the lowest at approximately $1.05M**. In terms of approved claims, **Dealer 014 recorded 140 approvals compared with 112 for Dealer 016**, while **Dealer 009 had the highest number of denied claims at 147 and Dealer 010 the lowest at 105**.

⏱️ Servicing Dealer Decision Processing

Operational efficiency also showed variation across servicing dealers. **Dealer 004 averaged approximately 80 days for claim decision processing**, with individual cases ranging from **1 to 150 days**, while **Dealer 010 averaged approximately 76 days**, with a range of **4 to 150 days**. The repeated **150-day maximum** suggests the presence of extended-processing cases that may warrant additional review.

💰 Selling Dealer Financial Exposure

The selling-dealer analysis showed another meaningful spread in financial exposure. **Selling Dealer 005 generated the highest total claim amount at approximately $1.45M**, whereas **Dealer 016 recorded the lowest at approximately $1.12M**, representing a difference of roughly **$330K**.

✅ Selling Dealer Claim Outcomes

Decision outcomes varied substantially across selling dealers. **Dealer 019 recorded the highest number of approved claims at 157**, while **Dealer 018 recorded only 10 approved claims**. On the denial side, **Dealer 013 recorded the highest number of denied claims at 139**, whereas **Dealer 019 recorded the lowest at 103**.

⏳ Selling Dealer Processing Efficiency

Selling-dealer processing times also highlighted operational variation. **Dealer 004 averaged approximately 80 days**, with a processing range of **1 to 150 days**, while **Dealer 020 averaged approximately 71 days**, with cases ranging from **4 to 148 days**.

🗓️ Year & Month Financial Trends

The time-based analysis revealed distinct financial highs and lows. **April 2021 recorded the highest monthly total claim amount at approximately $824,432**, representing the peak period for overall claim exposure. Meanwhile, **May 2021 recorded the highest average claim amount at approximately $5,445**, indicating that claims during that month were individually more expensive on average. At the other end of the spectrum, **February 2021 recorded both the lowest total claim amount at approximately $538,529 and the lowest average claim amount at approximately $4,682**.



🔹 THE STORYTELLING

📖The dashboard storytelling was intentionally designed to progress from **"What is happening?" to "Where is it happening?" to "What may be driving it?"**

- The dashboard tells a progressive business story by moving from **overall financial exposure to the factors that provide context behind that exposure**. The dataset represents approximately **$25.04M in total claim exposure**, with **Suspension contributing the highest total claim amount at about $5.20M**, while **Engine had the highest average claim severity at approximately $5,066**- showing that the category generating the most total dollars is not necessarily the category with the most expensive individual claims.
- At the vehicle level, **Honda carried the highest financial exposure at approximately $5.19M**, while **Kia recorded the highest average claim amount at approximately $5,108**, reinforcing the distinction between **claim volume and claim severity**. Mileage adds another layer, with the **50K–75K band generating approximately $8.49M**, compared with **$3.44M for vehicles in the 0–10K range**, suggesting that higher-mileage segments represent greater aggregate financial exposure, while recognizing that the data alone does not establish causation.
- The dealer analysis then shifts the story toward operational performance: **Servicing Dealer 009 had the highest total claim exposure at about $1.40M**, while Dealer 010 had the lowest at about $1.05M; approval and denial counts also varied across dealers, highlighting differences in claim outcomes and workload. Processing-time analysis adds an efficiency perspective, with **Servicing Dealer 004 averaging approximately 80 days** and Dealer 010 about **76 days**, while both reached lengthy maximum processing times, signaling potential opportunities for process review. On the selling side, **Dealer 005 recorded the highest financial exposure at approximately $1.45M**, compared with **$1.12M for Dealer 016**, while approval and denial patterns differed considerably across selling dealers.
- Finally, the time-series analysis reveals the strongest financial and severity fluctuations: **April 2021 recorded the highest total claim amount at approximately $824K**, **May 2021 recorded the highest average claim amount at about $5,445**, and **February 2021 recorded both the lowest total exposure at approximately $539K and the lowest average claim amount at approximately $4,682**.
- Together, these findings create a connected analytical narrative: **financial impact is shaped by claim category, vehicle characteristics, mileage, dealer behavior, operational efficiency, and time**, allowing the dashboard to move beyond simply reporting results toward identifying **where exposure is concentrated, where behavior differs, and where further business investigation would create the most value**.

⚠️ CHALLENGES & HOW THEY WERE ADDRESSED

One challenge was determining the appropriate **data types for identifiers versus measurable fields**. This was addressed by treating identifiers as text and numerical business measures as numbers, preventing inappropriate aggregation. Another challenge involved the **Agent Score field**, which was stored as descriptive text. Rather than replacing the original information, a separate numerical scoring field was created, preserving both **business meaning and analytical usability**. The dataset also required creation of a meaningful **mileage segmentation**, which was handled through Power Query conditional logic. During modeling, a conscious decision was made **not to over-engineer the data model**. Because this was a single 5,000-row fact table, only the dimensions providing meaningful analytical reuse were separated. Sorting and time-related fields also required attention, particularly for **Call Hour and Year Month**, ensuring that visuals followed chronological rather than alphabetical ordering.

💰 Understanding Financial Exposure vs. Claim Severity

One of the key analytical challenges was distinguishing between **overall financial exposure and the severity of individual claims**. The dataset contained approximately **$25.04M in total claim exposure**, but the category with the highest total cost was not necessarily the category with the highest average claim. For example, **Suspension generated the highest total exposure at approximately $5.20M**, while **Engine recorded the highest average claim amount at approximately $5,066**. This was addressed by analyzing **Total Claim Amount and Average Claim Amount as separate measures**, allowing the dashboard to distinguish between **high-frequency financial exposure and high-severity claims** rather than treating them as the same business problem.

🚗 Separating Vehicle Volume from Financial Impact

Another challenge was determining whether a vehicle make's financial exposure was driven by **claim frequency or claim severity**. **Honda recorded the highest total claim exposure at approximately $5.19M**, yet **Kia had the highest average claim amount at approximately $5,108**. This demonstrated that the vehicle with the highest overall financial impact was not necessarily the vehicle with the most expensive individual claims. To address this, the analysis combined **vehicle-level claim volume, total claim amount, and average claim amount**, providing a more balanced view of vehicle financial performance.

🛣️ Interpreting Mileage Without Assuming Causation

Mileage introduced another analytical challenge because **total mileage is an exposure measure, not a direct failure or risk rate**. The **50K–75K mileage band generated approximately $8.49M in total claim exposure**, compared with approximately **$3.44M for the 0–10K band**. Rather than

concluding that higher mileage directly causes higher claim costs, the analysis treated mileage as an important **contextual variable** and compared it with claim volume and financial exposure. This helped ensure that the findings remained **evidence-based and avoided unsupported causal conclusions**.

🏢 Comparing Servicing Dealers Across Multiple Dimensions

Dealer analysis presented the challenge of determining what actually represents stronger or weaker dealer performance. **Servicing Dealer 009 had the highest total claim exposure at approximately $1.40M**, while **Dealer 010 had the lowest at approximately $1.05M**. At the same time, approved and denied claim counts showed different patterns, with **Dealer 014 recording 140 approved claims**, while **Dealer 016 recorded 112**, and **Dealer 009 reaching 147 denied claims compared with 105 for Dealer 010**. Rather than evaluating dealers using a single ranking, the dashboard addressed this by examining **financial exposure, claim outcomes, and processing efficiency together**, creating a more complete dealer-performance perspective.

⏱️ Handling Wide Variations in Processing Time

Decision-processing time created another challenge because average values alone could hide significant case-level variation. **Servicing Dealer 004 averaged approximately 80 days**, with individual processing times ranging from **1 to 150 days**, while **Dealer 010 averaged approximately 76 days**, with a range of **4 to 150 days**. Similar variation appeared in the selling-dealer analysis, where **Dealer 004 averaged 80 days** and **Dealer 020 averaged approximately 71 days**, with maximum processing times of **150 and 148 days respectively**. To address this, the analysis considered **average, minimum, and maximum processing duration together**, making it possible to identify potential long-running cases without assuming that every extended duration represents inefficiency.

💰 Distinguishing Selling Dealer Exposure from Servicing Dealer Exposure

A further challenge was keeping **selling-dealer and servicing-dealer roles analytically distinct**. Selling Dealer 005 recorded the highest total claim exposure at approximately **$1.45M**, while Dealer 016 recorded approximately **$1.12M**, but claim outcomes and processing metrics showed different dealer rankings. This was addressed by analyzing **selling and servicing dealers separately**, preventing their roles from being blended together and allowing the dashboard to reveal where financial and operational patterns differ depending on the dealer relationship.

✅ Understanding Approval and Denial Differences

Approval and denial metrics also required careful interpretation because **claim counts and claim rates tell different stories**. For selling dealers, **Dealer 019 recorded 157 approved claims**, while Dealer 018 recorded only 10, and **Dealer 013 had 139 denied claims compared with 103 for Dealer 019**. These differences could reflect different claim volumes or claim compositions rather

than automatically indicating better or worse performance. The solution was to evaluate **approved and denied counts alongside approval/denial rates and total claims**, ensuring that the analysis considered both **scale and proportion**.

📅 Identifying Time-Based Financial Anomalies

The monthly analysis introduced the challenge of separating a **high-volume financial spike from a high-severity event**. **April 2021 recorded the highest total claim amount at approximately $824,432**, while **May 2021 recorded the highest average claim amount at approximately $5,445**. Conversely, **February 2021 had both the lowest total claim exposure at approximately $538,529 and the lowest average claim amount at approximately $4,682**. The dashboard addressed this by analyzing **total and average claim amounts together**, making it possible to determine whether a financial change appeared to be related to claim volume, claim severity, or both.



🌟 WHY THIS PROJECT MATTERS

This project demonstrates how raw operational data can be transformed into a **business-oriented analytical solution**. The real value is not simply that the dashboard contains KPIs. It demonstrates the ability to **clean imperfect business data, make modeling decisions, build reusable measures, select appropriate visualizations, investigate anomalies, recognize relationships, and translate findings into business language**.

The project also demonstrates an important Business Analyst mindset: **not every high number is a problem, not every low number is positive and not every relationship is causal**. The analyst's responsibility is to identify meaningful signals, provide context, and guide the next business question. That approach makes the dashboard useful to stakeholders across **claims management, finance, operations, dealer management, customer service, and warranty strategy**.



🎯 STRATEGIC RECOMMENDATIONS

💰 Optimize Financial Exposure by Claim Type

The analysis indicates that the claims portfolio carries approximately **$25.04M in total financial exposure**, with **Suspension claims contributing the highest total exposure at approximately $5.20M**, while **Engine claims have the highest average claim value at approximately $5,066**. This difference shows that **high total exposure and high individual claim severity are not necessarily driven by the same claim category**. Management should therefore track both **claim frequency and claim severity** when prioritizing cost-control initiatives, rather than focusing only on the category with the largest dollar total. A deeper review of high-value Engine claims and high-volume Suspension claims could help identify opportunities for **targeted warranty controls, preventive service strategies, and more effective claims monitoring**.

🚘 Strengthen Vehicle-Level Risk Monitoring

Honda represents the highest total vehicle-level claim exposure at approximately $5.19M, while **Kia records the highest average claim amount at approximately $5,108**. This difference highlights the importance of separating **overall financial burden from individual claim severity**. Rather than treating one vehicle make as the primary risk segment, management should evaluate vehicle performance through a combination of **claim volume, total exposure, average severity, mileage, and claim type**. This can help identify whether higher costs are associated with a larger number of claims or with more expensive individual cases.

🛣️ Use Mileage as a Proactive Risk Indicator

The **50K–75K mileage band generated the highest total claim exposure at approximately $8.49M**, compared with **$3.44M for vehicles in the 0–10K range**. This pattern suggests that **higher-mileage vehicles represent a greater concentration of aggregate claim costs** within the dataset. While the result does not establish that mileage directly causes higher claim costs, it provides a strong basis for proactive analysis. Management could use mileage bands in **warranty planning, preventive maintenance programs, risk segmentation, and future claims forecasting**, particularly for vehicles approaching higher-use ranges.

🏢 Introduce Dealer-Level Performance Benchmarking

The servicing-dealer analysis shows substantial differences in financial and decision outcomes. **Servicing Dealer 009 recorded approximately $1.40M in total claim exposure**, compared with **$1.05M for Dealer 010**, while approved and denied claim volumes also varied across dealers.

This suggests that dealer performance should be monitored using a **balanced scorecard rather than a single metric**. A practical benchmark should combine **total claim exposure, average claim amount, approval rate, denial rate, claim volume, and processing time** so that management can distinguish high workload from genuinely unusual behavior.

⏱️ Investigate Extended Decision-Processing Cycles

Processing times show that some claims can remain open for extended periods, with **Dealer 004 averaging approximately 80 days and reaching a maximum of 150 days**, while **Dealer 010 averaged approximately 76 days and also reached 150 days**. At selling-dealer level, **Dealer 004 also averaged approximately 80 days**, compared with **71 days for Dealer 020**. These results suggest an opportunity to investigate **long-running claims and process bottlenecks**, particularly cases approaching the upper end of the observed range. Reviewing documentation, escalation steps, claim complexity, and approval workflows could help reduce unnecessary delays while maintaining appropriate claims controls.

✅ Improve Decision Consistency Across Dealers

The analysis identified notable differences in decision outcomes, including **140 approved claims for Servicing Dealer 014 versus 112 for Dealer 016**, while Selling Dealer 019 recorded **157 approved claims compared with only 10 for Dealer 018**. Denial activity also varied, with **Servicing Dealer 009 recording 147 denied claims** and **Selling Dealer 013 recording 139**. These variations should be examined in context rather than interpreted as direct performance rankings. Management should compare dealer results against **claim volume and claim composition** to determine whether differences are driven by the types of cases submitted or by potentially inconsistent decision patterns.

🗓️ Use Peak Months for Exception-Based Review

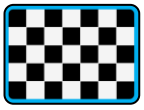
The time-series analysis identified **April 2021 as the highest month for total claim exposure at approximately $824,432**, while **May 2021 recorded the highest average claim amount at approximately $5,445**. Conversely, **February 2021 had both the lowest total claim amount at approximately $538,529 and the lowest average claim value at approximately $4,682**. These periods should be used as examples of **exception-based monitoring**, where unusually high or low periods trigger a deeper drill-down into claim type, vehicle make, mileage, dealer, and decision outcomes. This approach allows management to investigate meaningful deviations without reviewing every period with the same level of effort.

🚀 Build a More Predictive Claims Strategy

The current analysis provides a strong foundation for future enhancements such as **claim-cost forecasting, anomaly detection, dealer benchmarking, mileage-based risk scoring, and predictive claim analysis**. Historical patterns across **vehicle make, claim type, mileage, dealer, and decision outcomes** can be used to identify recurring behaviors and support more proactive planning. The long-term opportunity is to move from asking **"What happened?"** toward **"What is likely to happen next, and where should the business intervene?"**

🏆 Strategic Takeaway

Overall, the analysis suggests that the greatest value comes from treating **financial exposure, claim severity, vehicle usage, dealer behavior, and process efficiency as interconnected business dimensions**. The recommended strategy is to prioritize **high-exposure claim categories, higher-mileage segments, unusual dealer behavior, and extended processing cases**, while using benchmarks and exception-based monitoring to focus management attention where it matters most. This creates a more disciplined approach to **cost management, operational improvement, claims governance, and data-driven decision-making**.



CONCLUSION

🏆 Final Project Takeaway

This Power BI assessment demonstrates an end-to-end ability to transform a **5,000-record vehicle claims dataset into a structured, interactive and business-focused analytical solution**. The strongest aspect of the project is the progression from **data preparation → modeling → DAX → visualization → trend detection → anomaly investigation → business interpretation → strategic recommendation**.

Rather than stopping at "what happened," the analysis consistently moves toward **"where is the impact, what pattern is visible, what relationship may explain it, and what should the business investigate next?"**

That is the core value of the project: **using analytics not simply to report the business, but to help the business ask better questions and make more informed decisions.**



AI Usage Disclosure

ChatGPT was used as a supporting analytical and documentation tool throughout this project. AI assistance was used to help **identify appropriate data types for different fields, understand and validate relationships within the data model, and determine relevant performance indicators and core business measures** for the analysis. ChatGPT was also used to support the development and refinement of **calculated measures and DAX formulas**, including KPI calculations, comparison logic, rates, and dynamic analytical measures.

For the documentation and communication layer, AI assistance was used for **paraphrasing and refining analytical findings, improving professional wording, and translating technical observations into clear business-oriented insights**. It was also used to support the creation and structuring of the **Business Requirements Document (BRD)**, including project context, objectives, business questions, analytical requirements and professional project documentation. **All final analytical decisions, data validation, dashboard design choices, interpretation of results and implementation were reviewed and applied by me based on the project requirements and dataset.** AI was used as a productivity and guidance tool, while the final solution reflects my own analytical judgment and implementation.